

From Spec to Sign-off: Agentic NPI validation workflow transformation

Rahul Shah *Partner, Director of Software Engineering,* ***Microsoft***

Vikram Vemulapalli *VP, Hardware Test, Validation & Quality,* ***Microsoft***

OCP PLATINUM® Member

Agenda

01 The Problem

02 Methodology

03 Infrastructure

04 Demo, Lessons and Takeaways

05 Call to Action

The Problem

Long Runway with Heavy Touchpoints

Semi-automated, expensive touchpoints.

Spec Review

Subjective and manual spec reviews.

Test Coverage & Authoring

Manual test plan and test case authoring.

Semi-automated test cases.

Reusability & Traceability

Limited specs to test plan traceability.

Escapes, Debug & RCA

Semi-automated debugging and RCA.

Sign-off

Manual results review, subjective sign-offs.

Methodology

01 **Current Workflow**

02 **Domain Data Standardization**

03 **AI Guardrails**

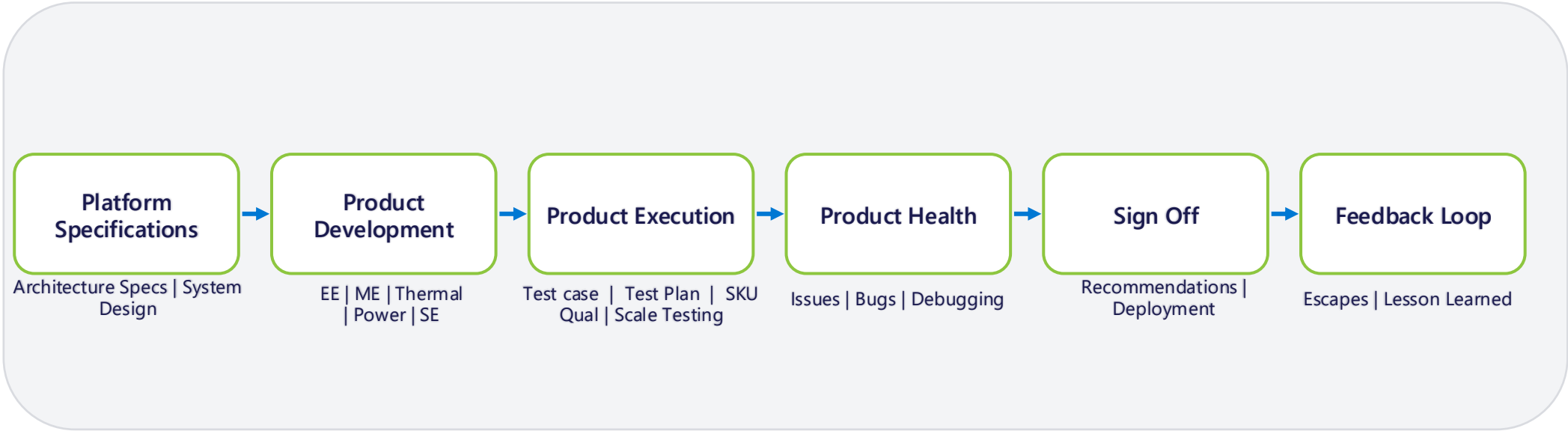
04 **Purpose built agents**

05 **Workflow transformation**

Methodology

01 Current Workflow

NPI Platform Validation



Methodology

02 Domain Data Standardization



Methodology

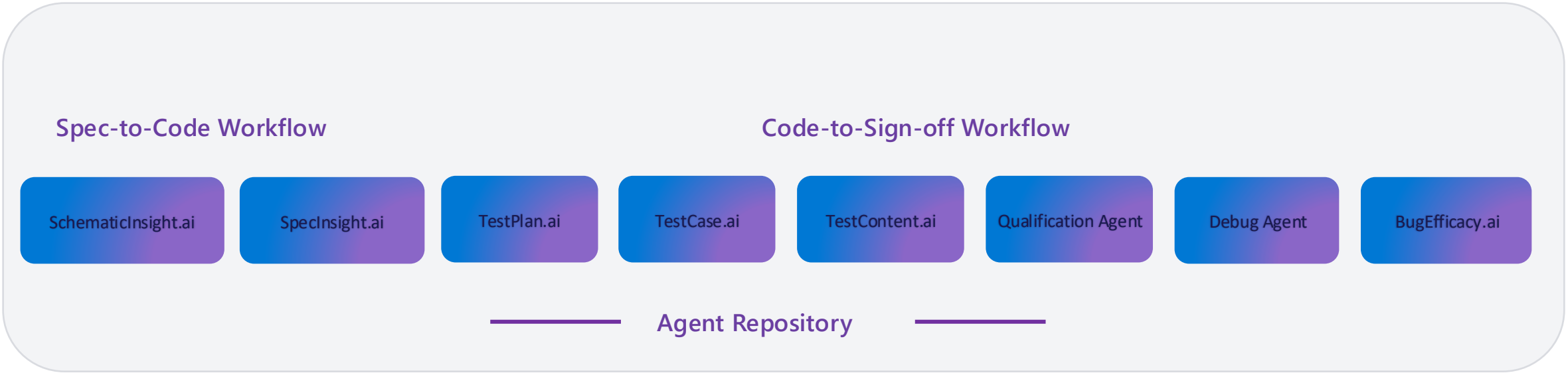
03 AI Guardrails

Compliance

- Agent Registration
- Responsible AI
- Security & Privacy
- Observability
- Livesite

Methodology

04 Purpose built agents



Methodology

05 **Workflow Transformation**

Domain Standardization

- Qualification
- Bug
- Digital Twin
- Testcase Library
- Incidents
- Lab Inventory

NPI Validation

Platform Specifications	Product Development	Product Execution	Product Health	Sign Off	Feedback Loop
Architecture Specs | System Design	EE | ME | Thermal | Power | SE	Test case | Test Plan | SKU Qual | Scale Testing	Issues | Bugs | Debugging	Recommendations | Deployment	Escapes | Lesson Learned

Spec-to-Code Workflow

Code-to-Sign-off Workflow

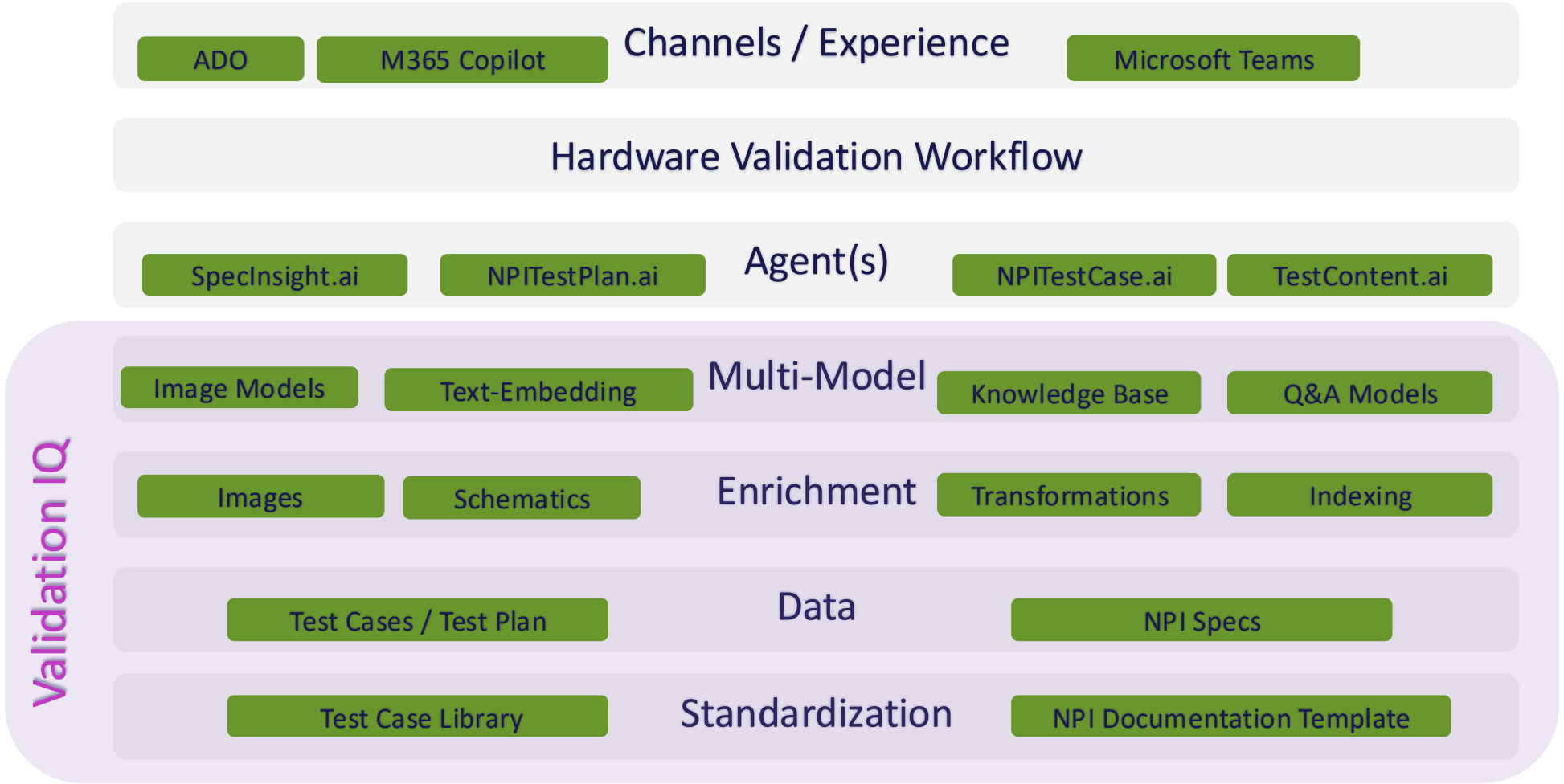
- SchematicInsight.ai
- SpecInsight.ai
- TestPlan.ai
- TestCase.ai
- TestContent.ai
- Qualification Agent
- Debug Agent
- BugEfficacy.ai

Agent Repository

Compliance

- Agent Registration
- Responsible AI
- Security & Privacy
- Observability
- Livesite

Infrastructure

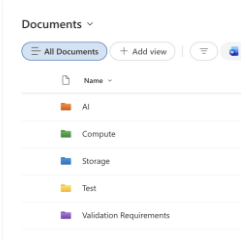


Demo: Spec to Sign-off, End to End

01 · INGEST

System of Record (SharePoint)

Spec and schematic documents land in SharePoint



02 · EXTRACT & ENRICH

Validation IQ

Extraction jobs build the knowledge base and vector data

03 · AUTHOR

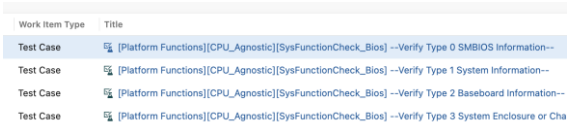
SpecInsight.ai + SchematicInsight.ai

Agents reason over the knowledge base to draft test plans and test cases in ADO

04 · APPROVE

Master Testcase Library (ADO)

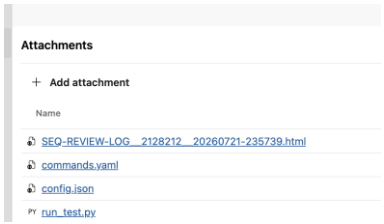
Engineers review and approve; approval triggers the next agent



05 · GENERATE

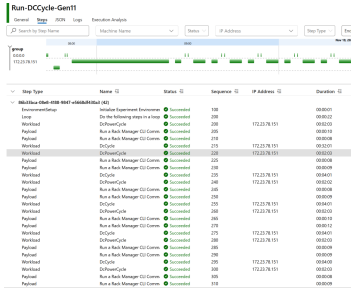
TestContent.ai

Writes executable test scripts for every approved test case



06 · EXECUTE

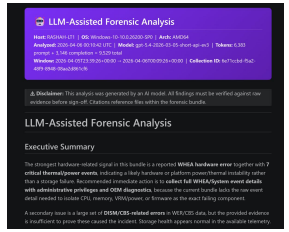
Automation



07 · ANALYZE

Debug Agent + Deep Lens

Reviews and analyzes results to surface failures and trends



08 · DECIDE

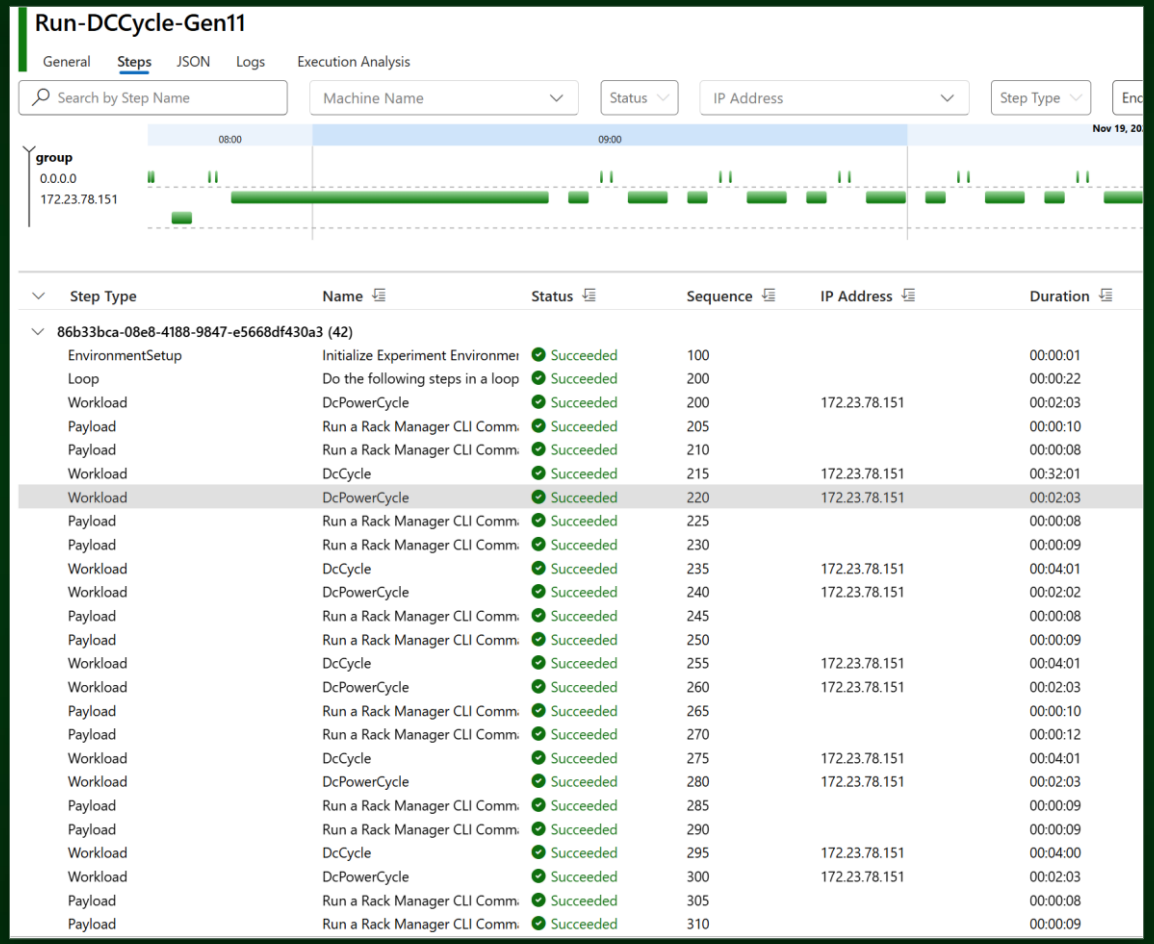
Sign-off or BugEfficacy.ai

Signs off the platform, or files actionable bugs

↺ **Feedback loop** Escapes, bugs and learnings flow back into the knowledge base and master testcase library, sharpening every next NPI cycle.

Demo: Automated Run to AI Forensic Analysis

06 · EXECUTE **End to End Test Case Execution Workflow**



Step Type	Name	Status	Sequence	IP Address	Duration
86b33bca-08e8-4188-9847-e5668df430a3 (42)					
EnvironmentSetup	Initialize Experiment Environmer	Succeeded	100		00:00:01
Loop	Do the following steps in a loop	Succeeded	200		00:00:22
Workload	DcPowerCycle	Succeeded	200	172.23.78.151	00:02:03
Payload	Run a Rack Manager CLI Comm	Succeeded	205		00:00:10
Payload	Run a Rack Manager CLI Comm	Succeeded	210		00:00:08
Workload	DcCycle	Succeeded	215	172.23.78.151	00:32:01
Workload	DcPowerCycle	Succeeded	220	172.23.78.151	00:02:03
Payload	Run a Rack Manager CLI Comm	Succeeded	225		00:00:08
Payload	Run a Rack Manager CLI Comm	Succeeded	230		00:00:09
Workload	DcCycle	Succeeded	235	172.23.78.151	00:04:01
Workload	DcPowerCycle	Succeeded	240	172.23.78.151	00:02:02
Payload	Run a Rack Manager CLI Comm	Succeeded	245		00:00:08
Payload	Run a Rack Manager CLI Comm	Succeeded	250		00:00:09
Workload	DcCycle	Succeeded	255	172.23.78.151	00:04:01
Workload	DcPowerCycle	Succeeded	260	172.23.78.151	00:02:03
Payload	Run a Rack Manager CLI Comm	Succeeded	265		00:00:10
Payload	Run a Rack Manager CLI Comm	Succeeded	270		00:00:12
Workload	DcCycle	Succeeded	275	172.23.78.151	00:04:01
Workload	DcPowerCycle	Succeeded	280	172.23.78.151	00:02:03
Payload	Run a Rack Manager CLI Comm	Succeeded	285		00:00:09
Payload	Run a Rack Manager CLI Comm	Succeeded	290		00:00:09
Workload	DcCycle	Succeeded	295	172.23.78.151	00:04:00
Workload	DcPowerCycle	Succeeded	300	172.23.78.151	00:02:03
Payload	Run a Rack Manager CLI Comm	Succeeded	305		00:00:08
Payload	Run a Rack Manager CLI Comm	Succeeded	310		00:00:09

07 · ANALYZE Deep Lens flags a WHEA error and 7 thermal/power events

LLM-Assisted Forensic Analysis

Host: RASHAH-LT1 | **OS:** Windows-10-10.0.26200-SP0 | **Arch:** AMD64
Analyzed: 2026-04-06 00:10:42 UTC | **Model:** gpt-5.4-2026-03-05-short-api-ev3 | **Tokens:** 6,383 prompt + 3,146 completion = 9,529 total
Window: 2026-04-05T23:39:26+00:00 → 2026-04-06T00:09:26+00:00 | **Collection ID:** 6e71ccbd-f5a2-48f9-8948-08aa2d861cf6

⚠ **Disclaimer:** This analysis was generated by an AI model. All findings must be verified against raw evidence before sign-off. Citations reference files within the forensic bundle.

LLM-Assisted Forensic Analysis

Executive Summary

The strongest hardware-related signal in this bundle is a reported **WHEA hardware error** together with **7 critical thermal/power events**, indicating a likely hardware or platform power/thermal instability rather than a storage failure. Recommended immediate action is to **collect full WHEA/System event details with administrative privileges and OEM diagnostics**, because the current bundle lacks the raw event detail needed to isolate CPU, memory, VRM/power, or firmware as the exact failing component.

A secondary issue is a large set of **DISM/CBS-related errors** in WER/CBS data, but the provided evidence is insufficient to prove these caused the incident. Storage health appears normal in the available telemetry.

Impact, Learnings and Takeaways

Impact

- Faster execution with better quality.
- Standardized master Test-case library leveraged across NPIs.
- 60% end to end turnaround time reduction.
- 75% reduction in sustaining qualifications turnaround.
- 100% automated test code development.

Learnings

- Domain data standardization comes first; agents are only as good as the schema beneath them.
- Guardrails ship with the agent: registration, responsible AI, security and observability from day one.
- Human-in-the-loop at sign-off keeps accountability with the validation owner.

Takeaways

- End to end workflow transformation
- Capability model with purpose built, specialized agents.
- Anchored in a single source of truth with standardized domains.
- Tighter hooks into existing bug, lab inventory and qualification tools.

Call to Action

Let’s collaborate

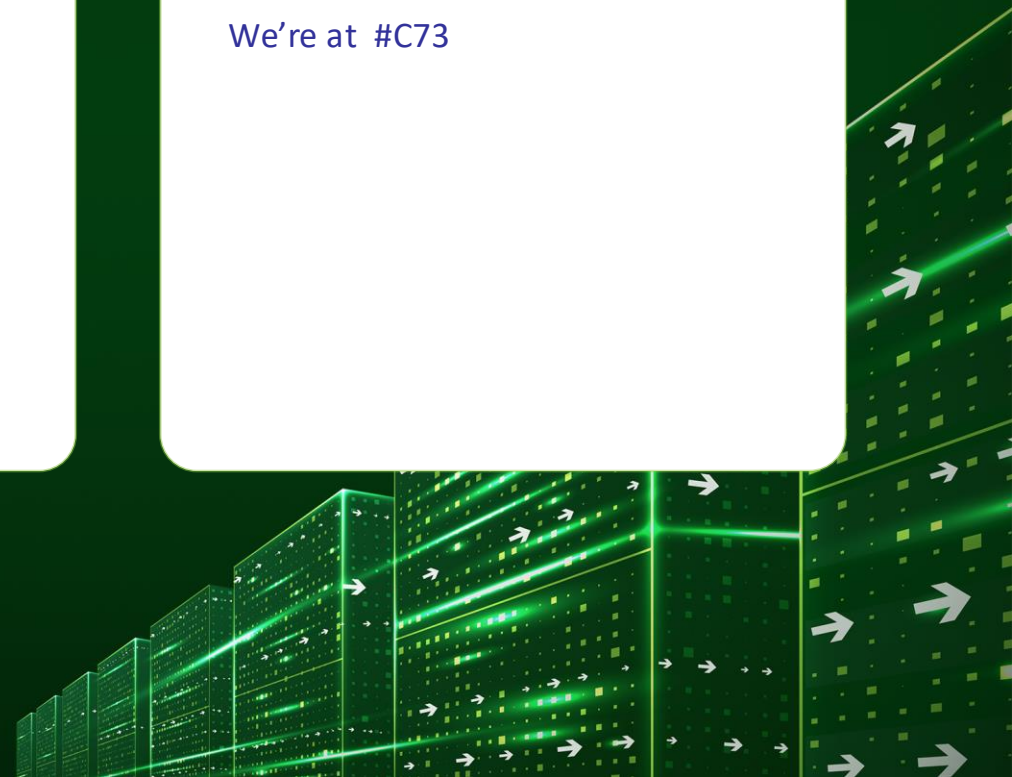
- Leverage OCP T&V spec schema for observability & telemetry.
- Leverage Testcases from OCP T&V.

Explore Virtual Client

- Open-source repository of industry standard benchmarks and workload evaluations.
- Part of OCP T&V Project.
- https://aka.ms/VirtualClient

Visit Microsoft Booth

We’re at #C73




Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California